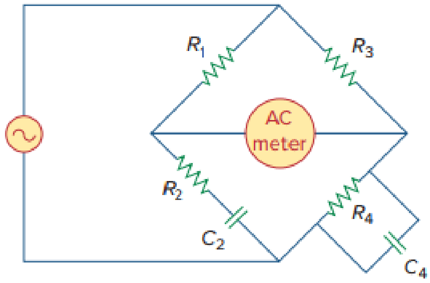
Chapter 9, Problem 85P

Problem

The ac bridge circuit of Fig. 9.85 is called a *Wien bridge*. It is used for measuring the frequency of a source. Show that when the bridge is balanced,

$$f = \frac{1}{2\pi\sqrt{R_2 R_4 C_2 C_4}}$$

Figure 9.85:



Step-by-step solution

Step 1 of 2

Let $Z_1 = R_1$, $Z_2 = R_2 + \frac{1}{j\omega_{C_2}}$, $Z_3 = R_3$

And $Z_4 = R_4 \parallel \frac{1}{j\omega_{C_4}}$

$$Z_4 = \frac{R_4}{j\omega R_4 C_4 + 1} = \frac{-jR_4}{\omega R_4 C_4 - j}$$

Since, $Z_4 = \left(\frac{Z_3}{Z_1} Z_2\right) \Rightarrow Z_1 Z_4 = Z_2 Z_3$

$$\frac{-jR_4 R_1}{\omega R_4 C_4 - j} = R_3\left(R_2 - \frac{j}{\omega_{C_2}}\right)$$

$$\frac{-jR_4 R_1(\omega R_4 C_4 + j)}{\omega^2 R_4^2 C_4^2 + 1} = R_3 R_2 - \frac{jR_3}{\omega_{C_2}}$$

Step 2 of 2

Equating the real and imaginary components,

$$\frac{R_1 R_4}{\omega^2 R_4^2 C_4^2 + 1} = R_2 R_3 \quad \rightarrow \text{ (1)}$$

$$\left(\frac{\omega R_1 R_4^2 C_4}{\omega^2 R_4^2 C_4^2 + 1}\right) = \frac{R_3}{\omega_{C_2}} \quad \rightarrow \text{ (2)}$$

Dividing (1) by (2),

$$\frac{1}{\omega R_4 C_4} = \omega R_2 C_2$$

$$\omega^2 = \frac{1}{R_2 C_2 R_4 C_4}$$

$$\omega^2 = 2\pi f = \frac{1}{\sqrt{R_2 C_2 R_4 C_4}}$$

$$f = \frac{1}{2\pi \sqrt{R_2 C_2 R_4 C_4}} \quad .$$